

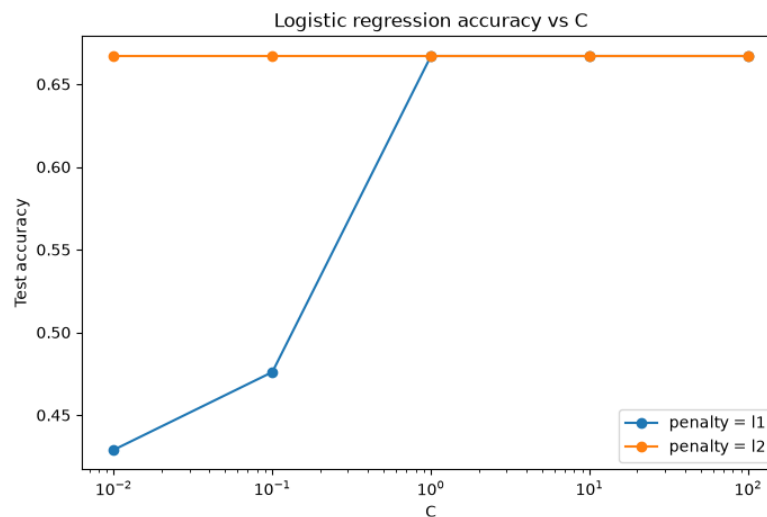
COVID-19 in China — Regression, Classification and Clustering

Analysing daily COVID-19 case trajectories in Chinese provinces through sliding-window regression, logistic classification, and k-means clustering.

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Overview

This project applies three families of machine-learning technique to daily confirmed COVID-19 case counts across Chinese provinces. A sliding-window approach converts Beijing's time series into a tabular regression and classification dataset. A separate clustering study tests whether geographic proximity explains province-level case similarity. The analysis reveals that outbreak magnitude, not physical distance, drives the structure of the data.



Results from the COVID-19 time-series modelling.

Goals

Demonstrate that standard regression, classification, and clustering methods can extract meaningful structure from a real epidemiological time series, and evaluate the limits of each technique on a small, imbalanced dataset.

Objectives

- Build a six-day sliding-window dataset from Beijing's daily case series and generate regression and classification targets.
- Fit linear regression and LASSO models to predict the sixth day from the previous five, and analyse how LASSO regularisation strength affects test MSE.
- Label each window by next-day direction (goup / godown / stable) and sweep logistic regression over L1/L2 penalty and C hyperparameters to study accuracy.

- Cluster all provinces two ways with k-means — by case series and by geographic coordinates — and compare partitions using the adjusted Rand index.
- Draw practical conclusions about the predictability of daily case direction and the relationship between geographic distance and outbreak similarity.

Approach

Data preparation uses a fixed-width sliding window to create supervised datasets from a univariate time series. Regression models are evaluated on held-out test MSE; the LASSO alpha sweep shows how coefficient shrinkage degrades predictive power. For classification, features are standardised before fitting logistic regression, and a grid over penalty type and C provides a structured hyperparameter analysis. Clustering uses k-means on two feature representations — normalised daily counts and latitude/longitude coordinates — with the adjusted Rand index quantifying agreement between the two partitions.

Outcomes

- Linear regression and LASSO both achieved low MSE on the Beijing regression dataset; LASSO MSE rose and plateaued as alpha increased beyond roughly 100.
- Classification accuracy was modest due to the small dataset size (67 windows) and class imbalance; moderate to large C values gave the best results for both L1 and L2 penalties.
- The adjusted Rand index between the geographic and case-based clusterings was near zero, showing the two partitions largely disagree.
- Hubei separated as its own cluster in the case-based grouping due to its outsized outbreak magnitude, while most other provinces clustered together.
- Geographic proximity is a poor predictor of COVID-19 case similarity; outbreak size is the dominant factor.